25. Monday, May 28. Inverses (6.5)

Question: Which subsets of $A \times B$ come from some function $A \to B$?

Example: $A = \{a_1, a_2, a_3\}$, $B = \{b_1, b_2\}$

Then $\{(a_1, b_2), (a_2, b_1)\}$ does not represent a function (Don't have output for a_3)

Also $\{(a_1, b_1), (a_2, b_1), (a_3, b_2), (a_3, b_1)\}$ does not represent a function

Because we have two outputs for the input a_3.

Theorem 1: Let A, B (non-empty) sets.

Suppose that a subset $f \subseteq A \times B$ satisfy the following properties:

(1) $\forall a \in A$, $\exists b \in B$ such that $(a,b) \in f$

(2) $\forall a \in A$, $\forall b, c \in C$, if $(a,b) \in f$ and $(a,c) \in f$, then $b = c$.

Then the rule $f(x) = y$ whenever $(x,y) \in f$ defines a function $A \longrightarrow B$.

Conversely, if this rule defines a function, then (1) and (2) hold.

Inverses (6.5)

Definition 2: Let $f: A \to B$ be a function and consider its graph representation:

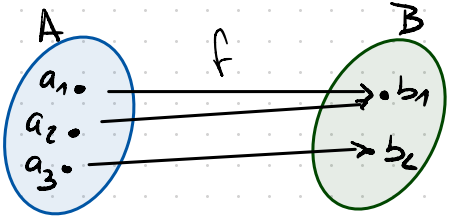
$$f = \{(a,b) \in A \times B \mid b = f(a)\} \subseteq A \times B$$

We define the **inverse of f**, denoted by f^{-1}, as the set

$$f^{-1} = \{(b,a) \in B \times A \mid b = f(a)\} = \{(b,a) \in B \times A \mid (a,b) \in f\}$$

May or may not be a function!

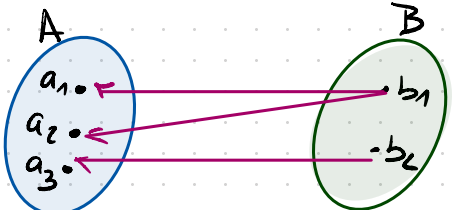
Example:



Here $f = \{(a_1,b_1), (a_2,b_1), (a_3,b_2)\} \subseteq A \times B$

So $f^{-1} = \{(b_1,a_1), (b_1,a_2), (b_2,a_3)\}$

Not a function! Two outputs for b_1.



Q: When is f^{-1} a function?

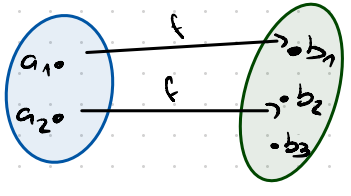
In the previous example, this failed because f is not injective:

$$f(a_1) = b_1 = f(a_2)$$

So we need f injective.

Q: Enough?

Example:



$$f = \{(a_1, b_1), (a_2, b_2)\}$$

$$f^{-1} = \{(b_1, a_1), (b_2, a_2)\}$$ Not a function! no output for b_3

Need f surjective.

Theorem 3: Let $f: A \to B$ be a function, seen as $f \subseteq A \times B$. Then

The inverse $f^{-1} \subseteq B \times A$ defines a function $B \to A$ if and only if f is bijective.

Proof: [$\Leftarrow$] Suppose $f: A \to B$ is a bijection. Consider $f^{-1} \subseteq B \times A$.

We show that $f^{-1} \subseteq B \times A$ satisfy both properties of Thm 1.

1) Let $b \in B$. Want to find $a \in A$ such that $(b,a) \in f^{-1}$

Since f is surjective, $\exists\, a \in A$ such that $f(a) = b$.
This means $(a,b) \in f$. Hence $(b,a) \in f^{-1}$.

2) Let $b \in B$. Suppose $\exists\, a_1, a_2 \in A$ such that $(b, a_1) \in f^{-1}$ and $(b, a_2) \in f^{-1}$. Want $a_1 = a_2$

Since $(b, a_1) \in f^{-1}$, we have $(a_1, b) \in f$, which means $f(a_1) = b$

Since $(b, a_2) \in f^{-1}$, we have $(a_2, b) \in f$, which means $f(a_2) = b$

Hence $f(a_1) = f(a_2)$. Since f is injective, we get $a_1 = a_2$, as desired.

So, by Thm 1, $f^{-1} \subseteq B \times A$ defines a function.

[$\Rightarrow$] (Sketch) If $f^{-1} \subseteq B \times A$ defines a function $B \to A$, Then

Property (1) in Thm 1 for f^{-1} says that f is surjective.

Property (2) in Thm 1 for f^{-1} says that f is injective.

□

Definition 4: If $f: A \to B$ is a bijection, we say it is an **invertible function**

In this case we showed $f^{-1} \in B \times A$ is a function, and we write $f^{-1}: B \to A$.

So we have proven:

Theorem 5: If $f: A \to B$ is a bijection, Then $f^{-1}: B \to A$ is a function defined by the rule: $f^{-1}(b) = a$. where $a \in A$ is the unique with $(a, b) \in f$.

Corollary 6: Suppose $f: A \to B$ is a bijection. Then $f^{-1}: B \to A$ satisfies:

(1) $\forall x \in A, \quad (f^{-1} \circ f)(x) = x$

(2) $\forall y \in B, \quad (f \circ f^{-1})(y) = y$

Proof: 1) Let $x \in A$ and write $y = f(x)$. Then

$$(x, y) \in f \Rightarrow (y, x) \in f^{-1} \Rightarrow x = f^{-1}(y)$$

Therefore $(f^{-1} \circ f)(x) = f^{-1}(f(x)) = f^{-1}(y) = x$. as claimed.

2) Exercise $\square$